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MILITARY SPECIFICATION

INSIGNIA, GRADE, ENLISTED PERSONNEL, METAL, ARMY AND AIR FORCE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers both the subdued and bright metal grade insignia used by Army enlisted personnel and metal grade insignia worn by Air Force enlisted personnel.

1.2 Classification. The insignia covered by this specification shall be as specified in the applicable military specification sheets (see 3.1, 6.2 and Supplement 1).

2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1 Specifications and standards. The following specifications and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

NN-P- 71 - Pallet, Material Handling, Wood, Stringer Construction 2-way and 4-way (partial)
QQ-Z-363 - Zinc Base Alloy, Die Casting
PPP-B-566 - Box, Folding, Paperboard
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-676 - Boxes, Setup

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, The Institute of Heraldry, US Army, Cameron Station, Alexandria, Va 22304-5050, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-F-495 - Finish, Chemical, Black, for Copper Alloys
MIL-P-15011 - Pallets, Material Handling, Wood post Construction, 4-way entry

(See supplement 1 for list of associated specifications)

STANDARDS

FEDERAL

FED-STD-151 - Metal; Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-2073-1 - DoD Materiel Procedures for Development and Application of Packaging Requirements.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

THE INSTITUTE OF HERALDRY, US ARMY

B-13-12 - Attaching Device, Heraldic Prong and Clutch Type

STANDARD

THE INSTITUTE OF HERALDRY, US ARMY

Standard Color Chip Set

(Figure 1 is a miniature reproduction of the referenced drawing and is attached for information only.)

(Copies of specifications, standards, and other Governments documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society for Testing and Materials (ASTM) Standards

ASTM-B-487 - Method for Measurement of Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section.

ASTM-D-3951 - Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. these documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 Detail requirements for individual insignia. Detail requirements or exceptions applicable to a particular insignia shall be as specified on the specification sheet for that insignia. In the event of conflict between the requirements of this specification and any specification sheet referenced herein, the specification sheet shall govern.

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3.2.1 Dimensions. All dimensions given as reference dimensions are information for bidding purposes only. The actual dimensions will be controlled by the Government furnished hub (see 6.2.1), except for army enlisted insignia of grade.

3.3 Standard sample. Unless otherwise specified (see 6.2), a standard sample of the applicable Air Force grade insignia shall be furnished for matching color and finish only. Variation from this specification may appear in the approved guide sample in which event the specification shall govern.

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.4).

3.5 Materials.

3.5.1 Copper base alloys. Copper base alloys shall be rolled polished, free from pits, scale (including red oxide), dents, nicks, scratches, cracks, segregations and foreign inclusions that will not be removed in later processing. When tested as specified in 4.4.1, the chemical composition of the applicable copper base alloy shall be as specified in table I.

TABLE I. Chemical composition of copper base alloy.

Alloy	Copper	Nickel 1/	Lead (max)	Iron (max)	Manganese (max)	Zinc	Others (max)
Yellow brass	64.0-68.5		.15	.05		Remainder	.15
Red brass	84.0-86.0		.05	.05		Remainder	.15
Nickel silver	63.0-66.5	16.5-19.5	.10	.25	.50	Remainder	
Gilding metal	94.0-96.0		.03	.05		Remainder	.13

1/ Cobalt counting as nickel.

3.5.2 Gold (for plating). Gold for plating shall be 24 karat.

3.5.3 Solder. The solder shall be a hard solder having a melting point of not less than 1075 degrees F.

3.5.4 Enamel.

3.5.4.1 Hard enamel. The hard enamel for Air Force insignia of grade shall be a hard fired vitreous type enamel fused with metallic oxides to produce the required color, shade and opacity.

3.5.4.2 Baking enamel. The enamel for Army subdued insignia of grade shall be a flat black synthetic baking enamel.

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3.5.5 Zinc aluminum alloy. The zinc aluminum alloy used for die casting shall conform to Alloy AG40A or AC41A of QQ-Z-363.

3.5.6 Lacquer. The lacquer shall be a pale clear synthetic lacquer. The use of pigmented lacquer shall not be permitted.

3.6 Design.

3.6.1 Army insignia. All Army insignia of grade shall be one piece except for the bright finished Specialist four which shall be two pieces. The eagle used for the Specialist four shall be an exact replica of a Government furnished hub (see 3.9) from which the contractor's working dies shall be extracted. A forcer may be used to bring up the design of the eagle for all subdued insignia. The impression of the forcer shall be regular and symmetrical with the design on the insignia front and the insignia back shall have no sharp or rough edges or corners.

3.6.2 Air Force insignia. The design of all Air Force insignia shall be an exact replica of the design on the Government loaned hubs (see 3.9), from which the contractor's working dies shall be extracted.

3.7 Construction.

3.7.1 Army insignia. Insignia shall be constructed in accordance with the applicable specification sheet and the requirements specified herein. Either stamping or die casting methods of manufacturing shall be acceptable for subdued insignia. When the die cast method is used, the insignia shall be free from porosity, blow holes or other casting defects. Prongs may be cast as an integral part of the insignia when die cast method is used. Before finishing, all flash lines shall be removed and the edges of the insignia shall be smooth. Unless otherwise specified on the applicable specification sheet, a tolerance of plus or minus .005 inch will be allowed on all interior dimensions. All bright insignia shall be stamped. When prongs are cast as an integral part of the insignia, the prong shall be tested according to paragraph 4.4.10 and no evidence of breaking or cracking shall occur.

3.7.2 Air Force insignia. All Air Force insignia shall be constructed in accordance with the applicable specification sheet and the requirements specified herein.

3.7.3 Stamping, trimming and piercing. Each insignia shall be struck in such a manner to insure a well defined die struck edge. Insignia shall be trimmed and when applicable pierced to the die struck outline. All edges shall be clean, smooth and free from burrs, drag, step, sharp or rough edges. The stamping, trimming and when applicable piercing operations shall not damage or distort the design or alter the shape of the insignia.

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3.7.4 Soldering and electronic fusion. At the option of the manufacturer, prongs for all insignia of grade, except for Army grades of Staff Sergeant and higher may be either soldered or electronically fused. For Army grades of Staff Sergeant and higher prongs shall be electronically fused. All soldering shall be accomplished using solder specified in 3.5.3. Soldered connection shall be completely joined without excess solder and all excess flux shall be removed. Soldered joint shall be clean, strong, smooth and free from burned or reduced areas. When tested as specified in 4.4.2, soldered joints shall not separate. When tested as specified in 4.4.3, prongs shall not separate at the fused joint. When the die cast method is used for prongs, testing shall be in accordance with 4.4.10 and no evidence of breaking shall occur.

3.7.5 Attaching device. Unless otherwise specified on the applicable specification sheet, attaching devices shall be prong and clutch type. The prongs and clutches shall conform to the details and dimensions specified in figure 1. The prongs shall be made from nickel silver specified in Table I unless die cast and shall be 5/16 inch plus or minus 1/32 inch in length. The clutches may be made from any type brass. When tested as specified in 4.4.4, it shall not be possible to remove the clutch by hand without first depressing the release wings.

3.7.6 Rivets. Rivets used for attaching superimposed device to the front of the base shall be solid or tubular and shall be made from any type brass. Rivets shall be securely peened or swaged to insure a tight and close fit between the component parts. At the option of the manufacturer, rivets may be soldered to, or be an integral part of the device.

3.8 Finish. Unless otherwise specified, the front and edges (surface visible from the front) shall be finished as specified on the applicable specification sheet and shall match the color of the standard TIOH color chip and finish of the standard sample (see 3.3). Where plating is specified, the entire insignia except for the attaching device shall be plated. The reverse and edges of all mirror-like insignia shall be bright. In addition to the finish specified on the applicable military specification sheet, all bright finish army insignia need only be clear lacquered on the front surface and edges as specified in paragraph 3.8.3. Complete lacquering of the entire insignia is optional. The pre-polished lacquered brass shall not be use for insignia requiring mirror-like polish.

3.8.1 Subdued insignia. The face and edges of all subdued insignia shall be coated with enamel specified in 3.5.4.2. Prior to enameling the entire insignia shall be given a black chemical finish conforming to MIL-F-495. When the subdued insignia is die cast from zinc aluminum specified in 3.5.5, the insignia shall be copper plated prior to finishing. When tested as specified in 4.4.7, the plating shall be a minimum of 0.0003 inch thick. When tested as specified in 4.4.5 and 4.4.6, the enamel shall not flake or chip and no color shall transfer to the cotton except for a slight trace of residue.

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3.8.2 Enameling. Unless otherwise specified on the applicable specification sheet. all enameling shall be accomplished using hard enamel. Enamel shall be applied so as to be uniform in color, free from bubbles, foreign inclusions, cracking, crazing or other defects which might affect appearance. There shall be no overrunning of enamel.

3.8.2.1 Colors. Enamel colors shall approximate the shades specified on the applicable specification sheet by chip numbers and shall match the standard (see 3.3) or approved (see 3.4) sample. The chip numbers specified are from The Institute of Heraldry, US Army Color Chip Set (see 2.1.2).

3.8.2.2 Hard enameling. Hard enamel specified in 3.5.4.1, shall be charged fired and unless otherwise specified, stoned level with the dikes. The enamel shall then be lapped, felted or wheel polished to produce a glass-like finish.

3.8.2.3 Baked enamel. Baked enamel shall be accomplished using enamel specified in 3.5.4.2.

3.8.3 Lacquering. The lacquer film shall be continuous, level, adherent and free from lint, dust and other foreign matter. When tested as specified in 4.4.8.1, there shall be no change in the appearance of the clear lacquer surface except a slight overall yellowing of the highlights and not more than two spots, each less than 1/16 inch largest dimension. When tested as specified in 4.4.8.2, the tissue paper shall not adhere to the clear lacquered surface.

3.8.4 Nickel plating. All plating shall be by electroplating methods. The plating shall be continuous and unbroken over the entire plated surface. There shall be no cut-through, shaded, peeled or blistered plating. Electroplating of attaching devices will not be required. The plating shall be not less than .00025 inch thick. Thickness shall be tested on the obverse face of the insignia.

3.8.5 Rhodium plating. Rhodium plating shall be continuous and adherent over the entire plated surface. There shall be no cut-through or areas of shaded, peeled or blistered plating. When tested as specify 4.4.9 no visible chemical reaction shall occur.

3.8.6 Gold plating. Brass insignia may be gold plated at the option of the manufacturer. If gold plated, the front and edges shall be plated and shall be continuous and unbroken over the entire plated surface. When tested as specified in 4.4.9, no visible chemical reaction shall occur.

3.9 Government loaned hubs. When a Government hub is furnished, (see 3.6.1 and 6.2) the hub shall be used for making the contractor's working dies necessary for one contract or order. The working dies shall be tooled and

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polished to remove any dents, nicks, scratches or other imperfections and any unornamented section shall be tooled and polished.

3.10 Marking. The contractor shall stamp his trademark or other identifying mark and the letters "GI" (for Government procurement only) legibly and inconspicuously on the back of each insignia. All other manufacture shall have the manufacturer's hallmark stamped on the reverse.

3.11 Workmanship. The finished item shall be clean, well made, and shall conform to the acceptable quality levels established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.3.1).

4.2.1 Certificate of compliance. Where certificate of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certificate.

4.3 First article inspection. Inspection and testing of the first article shall be made on a completely finished insignia for all provisions applicable to end product examination and tests.

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4.3.1 Quality conformance inspection. Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable test method document or applicable paragraph(s) in this specification.

4.3.3 Inspection for acceptance. Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.4 Inspection of components. In addition to the quality assurance provisions of the subsidiary specifications and drawings, inspection shall be performed on the materials listed in table II for the test characteristics shown therein. The inspection level shall be S-1 and the requirements shall be applicable to the lot average.

TABLE II. Testing of components.

Components and lots expressed in terms of	Characteristic	Rqm't para	Test method	#determ per s/u	Results reported	Sample unit
Copper base alloy as applicable (lb)	Chemical composition	3.5.1	Method 111 FED-STD-151	2 each element	Nearest 0.1%	One 4 oz. pc or equivalent for each alloy

4.3.4.1 Certification of compliance. Materials listed below may be accepted on the basis of a contractor's certification of compliance for requirements specified in the applicable paragraphs of this specification.

Component	Characteristics	Rqm't para
Baking enamel	Material identification	3.5.4.2
Hard enamel	Material identification	3.5.4.1
Lacquer	Material identification	3.5.6
Rhodium plating	Material identification	3.8.5
Nickel plating	Material identification	3.8.4

4.3.5 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material for which in-process

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inspection has indicated non-conformance. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specification requirements.

Requirement, operation or assembly	Characteristic	Rqm't para
Black chemical finish	Used as a base	3.8.1

4.3.6.1 Testing of the intermediate product. Insignia shall be tested for copper plating thickness in accordance with 4.4.7. There shall be two determinations per sample unit and the results shall be reported to the nearest 0.0001 inch. The sample unit shall be one plated insignia (from each plating batch) of one type. The thickness of nickel plated insignia shall be tested in accordance with standard commercial practice. In the event of dispute, the test method shall be ASTM-B-487. There shall be one determination per sample unit and results shall be reported as pass or fail. The sample unit shall be one insignia. The requirement is applicable to the individual unit, the inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

4.3.7 Inspection of end item.

4.3.7.1 Visual examination of insignia. Examination shall be made at a distance of 16 to 22 inches under normal daylight or its equivalent. The defects found during the examination shall be classified in accordance with 4.3.7.1.1 and 4.3.7.1.2. The lots shall be inspected in accordance with 4.3.7.1.3. The unit of product for these examinations shall be one completely fabricated insignia.

4.3.7.1.1 Examination for defects in finish, design, material, construction, workmanship and marking. Defects designated by an asterisk (*) shall be classified as major when seriously affecting appearance and serviceability and minor when affecting appearance and serviceability but not seriously.

Examine	Defects	Classification Major	(*)	Minor
Color and finish	Does not compare favorably with the approved guide sample	x	-	-
	Surfaces not smooth	-	*	-
	Discoloration or stain clearly noticeable . .	-	*	-
	Abraded, scuffed, scratched or otherwise damaged.	-	*	-

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Examine	Defects	Classification Major	(*)	Minor
Lacquering (bright only)	Lacquer coating hazy, cloudy or powdering . .	-	-	x
	Foreign matter embedded in finish, clearly noticeable	-	-	x
	Lacquer not smooth, not continuous or not adherent, i.e., flaking, blistering, rippling, peeling or has run, sag or drops	-	*	-
	Lacquer not set to touch.	-	-	x
Chemical finish	Not chemically finished (subdued only). . . .	x	-	-
	Not type of chemical finish specified (subdued only)	-	-	x
	Chemical finish is not continuous over the entire insignia (subdued only)	x	-	-
Enameling (when applicable)	Not enameled or area of no film	x	-	-
	Area of thin film	-	-	x
	Enamel not type specified	x	-	-
	Not adherent, i.e., blistered, flaking or peeling.	-	*	-
	Wrong color or not uniform.	x	-	-
	Wrong shade	-	*	-
	Not stoned smooth (when applicable)	-	*	-
	Is not level with dikes, e.g., contains high or low spots	-	*	-
	Cracking or crazing	x	-	-
	Hole, blister or not free of bubbles.	-	*	-
	Laps, overruns or skips	-	*	-
	Foreign matter embedded	-	*	-
	Discoloration clearly noticeable.	-	*	-
	Coating not continuous and level, i.e., runs, drips or drops	-	*	-
Design	Detail altered and does not conform to Government hub or pattern (when applicable).	x	-	-
	Any warp, twist or distortion producing irregular surface contour or outline	-	*	-
	Detail struck over, resulting in a double impression	x	-	-
Material	Not fabricated of the specified material. . .	x	-	-
Construction and workmanship	Not trimmed to die struck edge.	-	*	-
	Not pierced where required.	x	-	-
	Piercing not clean and smooth, e.g., any burrs, cutter, drill or file marks.	-	*	-
	Any component missing, i.e., prong, clutch, device	x	-	-

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Examine	Defects	Classification Major	(*)	Minor
	Superimposed device is not securely riveted to the base (when applicable), i.e., rivets are loose, poorly or insufficiently peened or swaged	x	-	-
	Riveted device is not closely fitted to the contour of the base.	-	*	-
	Superimposed device is not centered or positioned as required	x	-	-
	Impression of forcer (when used) not regular and symmetrical with the design.	-	*	-
	Dent, pit, dig, gouge, nick, pores, metal mark or mutilation	-	*	-
	Edges of forced areas not clean or smooth	-	-	x
	Distorted, bent, malformed or otherwise impaired	-	*	-
	Porosity, blow hole or other casting defect (when applicable).	-	*	-
	Burr, fin, rough or sharp edges, drag, step or flash	-	*	-
	Metal cracked or fractured.	x	-	-
	Any solder spattered on front or visible edge exposed or concealed by finish.	-	*	-
	Defective, e.g., any part damaged or malformed affecting use.	-	*	-
Attaching device	Not fused when required	x	-	-
	Not as specified.	-	*	-
	Prong not completely joined by solder	-	*	-
	Any area burned or reduced in soldering	-	*	-
	Flux or excess solder not removed	-	*	-
Marking	Not marked as specified	-	-	x
	Not in the location specified	-	-	x

4.3.7.1.2 Examination of insignia for defects in dimensions. Any dimension that is not within the specified tolerance shall be classified a defect.

4.3.7.1.3 Inspection levels and AQL's. The inspection levels and the acceptable quality levels (AQL's) expressed in defects per hundred units shall be as follows:

	Inspection level	AQL Major	AQL Total
For defects applicable to 4.3.7.1.1	II	2.5	15.0
For defects applicable to 4.3.7.1.2	S-2	(one class)	4.0

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4.3.7.2 Testing of the end item. Testing of the completely fabricated insignia shall be performed in accordance with table III for the characteristics shown therein.

TABLE III. End item testing.

Characteristic	Requirement paragraph	Test method	# Det. per sample unit
Test for lacquer:			
Liver of sulfur	3.8.3	4.4.8.1	1
Tackiness	3.8.3	4.4.8.2	1
Acid test for rhodium plating	3.8.5	4.4.9	1
Test for soldered joints	3.7.4	4.4.2	2
Test for fused joints	3.7.4	4.4.3	2
Knife test for baked enamel	3.8.1	4.4.5	1
Acetone test for baked enamel	3.8.1	4.4.6	1
Test for clutches	3.7.5	4.4.4	2
Test for integral prongs	3.7.1	4.4.10	8*

* Insignia are to be tested per sample unit as two prongs per insignia.

4.3.8 Examination for count of insignia in intermediate containers. Insignia packaged for shipment shall be examined to determine conformance with package markings and specified quantity. The sample unit for this examination shall be one box (interior package). Any box containing less than the specified or marked quantity of insignia shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units. The lot size shall be the number of intermediate containers.

4.3.9 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking requirements are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Markings (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing.
	Any component damaged, affecting serviceability.

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Workmanship	Inadequate application of components, such as, incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Weight or content (exterior and interior)	Number of intermediate containers is more or less than required; gross/net weight exceeds requirements.

4.3.10 Palletization examination. An examination shall be made to determine that the palletization is in compliance with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.4 Tests.

4.4.1 Chemical composition test. Chemical composition shall be determined in accordance with Method III of FED-STD-151. Results shall be evaluated to determine compliance with the requirements in 3.5.1.

4.4.2 Tests for soldered joints. Samples of insignia to be tested shall be placed in an oven maintained at 1065 degrees F., plus or minus 5 degrees F., for not less than 15 minutes. While at the specified temperature, the insignia shall be lifted by the smaller part. The sample shall then be examined to determine compliance with 3.7.4.

4.4.3 Test for electronically fused joints. Item to be tested shall be anchored on a horizontal surface. Each prong shall be grasped at least 1/3 its length above the fused joint and bent through an angle of 90 degrees (45

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degrees to each side of the vertical) until separation occurs. The fused joint shall then be examined to determine compliance with 3.7.4. A bending tool in the form of a 45 degree template may be used for this test provided the prong is gripped at least 1/3 its length above the fused joint.

4.4.4 Test for clutches. The clutch shall be removed and replaced ten times from the prong. An attempt shall then be made to remove the clutch by hand without first depressing the release wings, during which time the clutch and prong shall be inspected to determine compliance with 3.7.5.

4.4.5 Test for baked enamel. Draw a knife edge across the face side of the specimen making four parallel scores one-sixteenth inch apart. Repeat this process, drawing the knife edge this time perpendicular to the previously scored lines, making small blocks in the finish. The finish shall then be examined to determine compliance with 3.8.1.

4.4.6 Acetone test for baked enamel. The enameled surface shall be wiped five times with a piece of cotton saturated with acetone. The cotton shall be examined to determine compliance with 3.8.1.

4.4.7 Direct micro-thickness test for copper plating. Copper plating thickness shall be determined in accordance with method 521 of FED-STD-151. Results shall be evaluated to determine compliance with 3.8.1.

4.4.8 Lacquer tests.

4.4.8.1 Liver of sulfur test. Insignia shall be immersed in a 2 percent by weight CP liver of sulfur (sulfated potash) water solution, at a temperature of 100 degrees F., plus or minus 10 degrees F., for a period of 3 minutes. The insignia shall then be removed and rinsed in warm, cold, then hot water. The insignia shall then be wiped gently with an absorbent cellulose material or whirled to remove residual moisture and shall be air dried at room temperature (60 to 80 degrees F.) for 1 hour. The insignia shall then be inspected to determine compliance with the requirements specified in 3.8.3.

4.4.8.2 Tackiness test. At room temperature (60 to 80 degrees F.) press a piece of tissue paper against the lacquered surface for 15 seconds, using any pressure capable of being exerted between thumb and two fingers, after which, the pressure shall be released and the insignia inspected to determine compliance with the requirements specified in 3.8.3.

4.4.9 Continuity of gold, nickel and rhodium plating. Insignia to be tested for compliance with 3.7.5 may be unlacquered or have the lacquer completely removed with an organic solvent. The acid test is conducted as follows: using a test acid consisting of a solution with 50 percent by volume nitric acid (specific gravity 1.42) an equal volume at room temperature, place a drop of

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acid (each drop to be 1/8 inch minimum diameter) on three different spots on reverse side of the insignia. The drops shall remain for not less than one minute. A minimum of two of these spots shall withstand this test.

4.4.10 Test for integral prongs. Insignia to be tested will be placed in a jig which allows no movement of the insignia and positions prongs on a 30 degree angle above the horizontal. A vertical 10 lb. load shall be applied to the top third position of each prong. The prongs will be examined to determine compliance with 3.7.1.

5. Packaging

5.1 Preservation. Preservation shall be level A, C or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging.

5.1.1.1.1 Army insignia. Twelve insignia of one grade and finish only for Private, Private First Class, Corporal, Sergeant or Specialist Four, insignia of one grade only for the remaining grades shall be equally spaced on a 4-1/2 inch plus or minus 1/2 inch by 3-1/2 plus or minus 1/4 inch, white sulfite card a minimum of 0.011 inch thick and wrapped in not less than three thicknesses of anti-tarnish paper having a minimum basis weight of not less than 12 pounds per ream or inserted into a snug-fitting polyethylene film bag. The polyethylene film shall be a minimum of .002 inch thick and all seams and closures shall be affected by heat sealing. Prior to or during closure excess air within the bag shall be expelled.

5.1.1.1.2 Air Force insignia. Two insignia of one grade only shall be mounted on a 2 inch by 2-3/4 inch white sulfite card a minimum of 0.011 inch thick and wrapped in not less than three thicknesses of anti-tarnish paper having a minimum basis weight of not less than 12 pounds per ream or inserted into a snug-fitting polyethylene film bag. The polyethylene film shall be a minimum of .002 inch thick and all seams and closures shall be affected by heat sealing. Prior to or during closure, excess air within the bag shall be expelled.

5.1.1.2 Intermediate package.

5.1.1.2.1 Army insignia. Twenty unit packages of one grade and finish only shall be packaged in a snug-fitting paperboard box conforming to variety 1, style XIA of PPP-B-566. Closure shall be in accordance with the appendix of the container specification.

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5.1.1.2.2 Air Force insignia. Sixty unit packages of one grade only shall be packaged in a snug-fitting paperboard box conforming to variety 1, style XIA of PPP-B-566. Closure shall be in accordance with the appendix of the container specification.

5.1.2 Level C. Insignia of one grade and finish only shall be packaged in accordance with the applicable requirements of MIL-STD-2073-1.

5.1.3 Commercial. Insignia shall be preserved and packaged in accordance with the applicable requirements of ASTM-D-3951.

5.2 Packing. Packing shall be level A, B, C or Commercial as specified (see 6.2).

5.2.1 Level A. Insignia of one grade and finish only, packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to class weather resistant, V3s or V3c of PPP-B-636. Closure shall be in accordance with the appendix of the applicable specification. The weight of the contents of each shipping container shall not exceed 65 pounds.

5.2.2 Level B. Insignia of one grade and finish only, packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to type CF or SF, class domestic, grade 275 of PPP-B-636. Closure shall be in accordance with the appendix of the container specification. Weight of contents of each shipping container shall not exceed 65 pounds.

5.2.3 Level C. Insignia, packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of MIL-STD-2073-1.

5.2.4 Commercial. Insignia shall be preserved in accordance with the applicable requirements of ASTM-D-3951.

5.3 Palletization. Items packed as specified shall be palletized on a 4-way entry pallet in accordance with load type IA of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance bonding means C, K, L, O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, type I, style 1, size A, and fabricated from class and wood group specified, of MIL-P-15011, or 4-way, type IV; type V, class 1, size 2; or type VIII, fabricated from wood groups I, II, III or IV, grade A of NN-P-71. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking. In addition to any special markings required by the contract or purchase order, each unit pack, exterior container and palletized unit load shall be marked in accordance with MIL-STD-129 or ASTM-D-3951, as applicable.

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6. NOTES

6.1 Intended use. Insignia covered by this specification are intended to be worn by enlisted personnel of the Army and Air Force on designated uniforms to indicate the grade of the wearer.

6.2 Ordering data.

6.2.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Title, number and date of the applicable specification sheet (see 2.1 and supplement 1).
c. When a standard sample is not provided (see 3.3).
d. When a first article is not required (see 3.4).
e. When a Government hub will not be furnished (see 3.9).
f. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
g. When palletization is required (see 5.3).

6.3 Detail requirements. This specification should be used in conjunction with the applicable specification sheets which lists the detail requirements of the insignia to be procured (see 3.1).

6.4 First article. When a first article inspection is required, the item should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.3. The first article should consist of one completed insignia set. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Government loaned property. The contracting officer should arrange to loan the property listed in 3.10.

6.6 Samples. For access to guide samples, address the contracting activity issuing the invitation for bids.

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6.7 Subject term (key word) listing.

Electronic fusion
Emblem
Grade, enlisted personnel
Insignia, Enlisted personnel
Metal
Military specification
Rank
Rivets
Stamping

6.8 Changes from previous issue. Asterisks or vertical line are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - IH
Air Force - 11

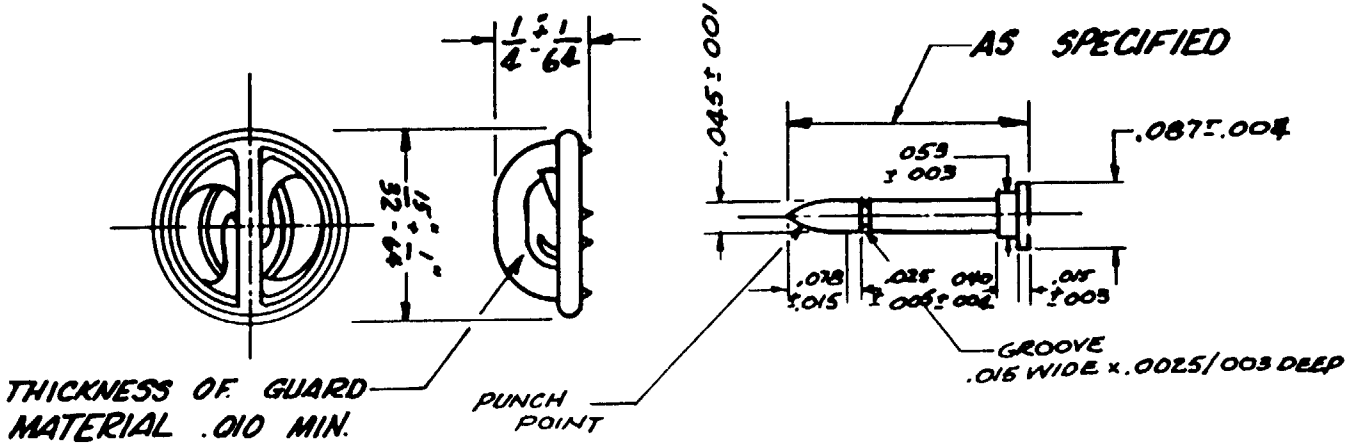
Review activities:
Air Force - 82, 99
DLA - CT

User activities:
Air Force - 45

Preparing activity:
Army - IH

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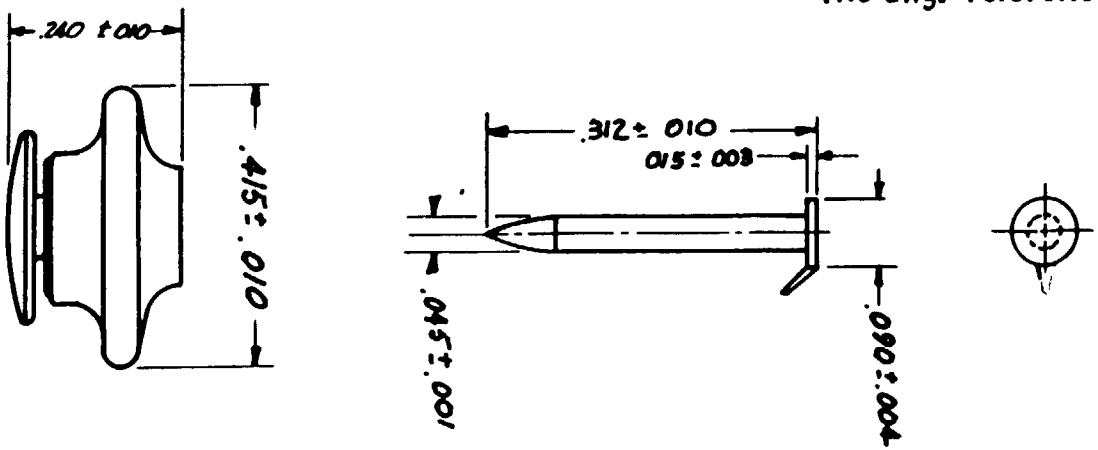
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CLUTCH PRONG

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The dwg. referenced in para. 2.1 governs.

FIGURE 1



FLAT BALL TYPE CLUTCH PRONG WITH PICK

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-I-14639D	Insignia, Grade, Enlisted Personnel, Metal, Army and Air Force, General Specification for

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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